

Java OOP Advanced Activity: Robot Factory

1. Folder Structure



2. Code Screenshots

Robot.java

```
JAVA > RobotFactory > factory > base > Robot.java > ...
1  package RobotFactory.factory.base;
2
3  public class Robot {
4      protected int energy;
5
6      public Robot() {
7          this.energy = 100;
8      }
9
10     public Robot(int energy) {
11         this.energy = energy;
12     }
13
14     public int displayEnergy() {
15         System.out.println("Current energy level: " + this.energy);
16         return this.energy;
17     }
18 }
```

SecurityBot.java

```
JAVA > RobotFactory > factory > types > SecurityBot.java > SecurityBot
1  package RobotFactory.factory.types;
2
3  import RobotFactory.factory.base.Robot;
4
5  public class SecurityBot extends Robot {
6
7      public SecurityBot() {
8          super(energy: 200);
9      }
10
11     public void patrol() {
12         this.energy -= 20;
13         System.out.println(x: "SecurityBot is patrolling the area. Energy -20.");
14     }
15
16     public void soundAlarm() {
17         this.energy -= 30;
18         System.out.println(x: "SecurityBot triggered the alarm! Energy -30.");
19     }
20
21     public void recharge() {
22         this.energy += 40;
23         System.out.println(x: "SecurityBot has recharged its power cells. Energy +40.");
24     }
25 }
```

WorkerBot.java

```
JAVA > RobotFactory > factory > types > WorkerBot.java > WorkerBot > chargeBattery()
1  package RobotFactory.factory.types;
2
3  import RobotFactory.factory.base.Robot;
4
5  public class WorkerBot extends Robot {
6
7      public void liftCrate() {
8          this.energy -= 15;
9          System.out.println(x: "WorkerBot lifted a heavy crate. Energy -15.");
10     }
11
12     public void chargeBattery() {
13         this.energy += 20;
14         System.out.println(x: "WorkerBot is recharging its battery. Energy +20.");
15     }
16
17     public void assemblePart() {
18         this.energy -= 10;
19         System.out.println(x: "WorkerBot assembled a part. Energy -10.");
20     }
21 }
```

RobotTest.java

```
JAVA > RobotFactory > RobotTest.java > RobotTest > main(String[])
1  package RobotFactory;
2
3  import RobotFactory.factory.types.WorkerBot;
4  import RobotFactory.factory.types.SecurityBot;
5
6  public class RobotTest {
    Run | Debug
7      public static void main(String[] args) {
8
9          /* WorkerBot Test */
10         System.out.println(x: "===== WorkerBot Test =====");
11         WorkerBot worker = new WorkerBot();
12
13         worker.liftCrate();
14         worker.liftCrate();
15
16         worker.assemblePart();
17         worker.assemblePart();
18
19         worker.chargeBattery();
20
21         System.out.println();
22         System.out.print(s: "WorkerBot - ");
23         worker.displayEnergy();
24
25         /* SecurityBot Test */
26         System.out.println();
27         System.out.println(x: "===== SecurityBot Test =====");
28         SecurityBot guard = new SecurityBot();
29
30         guard.patrol();
31         guard.patrol();
32
33         guard.soundAlarm();
34
35         guard.recharge();
36
37         System.out.println();
38         System.out.print(s: "SecurityBot - ");
39         guard.displayEnergy();
40     }
41 }
42
```

Sample Output

```
===== WorkerBot Test =====
WorkerBot lifted a heavy crate. Energy -15.
WorkerBot lifted a heavy crate. Energy -15.
WorkerBot assembled a part. Energy -10.
WorkerBot assembled a part. Energy -10.
WorkerBot is recharging its battery. Energy +20.

WorkerBot - Current energy level: 70

===== SecurityBot Test =====
SecurityBot is patrolling the area. Energy -20.
SecurityBot is patrolling the area. Energy -20.
SecurityBot triggered the alarm! Energy -30.
SecurityBot has recharged its power cells. Energy +40.

SecurityBot - Current energy level: 170
PS C:\Users\louis\Git\My_First_Repo>
```